Fair Division of Indivisible Goods

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Outline

Agents and Goods

Basic Setup

- **Agents:** $N = \{1, 2, ..., n\}$
- **Goods:** $M = \{1, 2, ..., m\}$
- Goods are **indivisible**

Valuation Function

It is a mathematical function that assign a mathematical value to a set of item .
Each agent $i \in N$ has a valuation function-
$v_i : 2^M \to \mathbb{R}_{\geq 0}$ for each agent i

- **Normalized:** $v_i(\emptyset) = 0$
- **Monotonic:** If $S \subseteq T$ then $v_i(S) \leq v_i(T)$

Addition Valuation

Definition (Additive Valuation)

A valuation is additive if for any bundle $S \subseteq M$:

$$v_i(S) = \sum_{g \in S} v_i(g)$$

where $v_i(g)$ is agent i's value for item g

Example

- Agent 1 values: $v_{11} = 3, v_{12} = 2, v_{13} = 1$
- For bundle $S = \{1, 3\}$: $v_1(S) = 3 + 1 = 4$

Fair Allocation

Definition

An allocation is fair if no one envies what someone else got.

Disjoint Condition

$A_i \cap A_j = \emptyset$ for all $i \neq j$
Each item allocated to at most one agent

Full Allocation

$\bigcup_{i \in N} A_i = M$
All items must be allocated

Partial Allocation

$\bigcup_{i \in N} A_i \subseteq M$
Some items may remain unallocated

Envy-Freeness

Definition (Envy-Free Allocation)

An allocation A is envy-free if:

$$v_i(A_i) \geq v_i(A_j) \quad \forall i, j \in N$$

- Each agent are satisfied and does not envy to others.

Example

Two agents, two items:

Agent	g_1	g_2
1	3	1
2	1	3

EF allocation:
$A_1 = \{g_1\}, A_2 = \{g_2\}$
Agent 1: value $= 3$
Agent 2: value $= 3$
No envy!

Proportionality

Definition (Proportional Allocation)

An allocation A is proportional if:

$$v_i(A_i) \geq \frac{1}{n} \cdot v_i(M) \quad \forall i \in N$$

Example

Three agents, three items (each valued at 2, 1, 1):

- Total value for each agent: $v_i(M) = 4$
- Proportional share: $\frac{4}{3} = 1.33$
- If each agent gets one item:
  - Agent getting item valued at 2: Satisfied
  - Agents getting items valued at 1: Not satisfied!

Impossibility of EF and Proportional Allocations

Impossibility Result

when the $n(N) > n(M)$ then EF and PROP are not exist.
$n(N)$ = Number of agents
$n(M)$ = Number of goods

Example

Simple case: 2 agents, 1 good

- Both value the good at 1
- Any allocation:
  - One agent gets value 1
  - One agent gets value 0
- Result:
  - Not envy-free (agent with 0 envies)
  - Not proportional (one gets less than $\frac{1}{2}$)

EF1 Definition

Definition (Envy-Free up to One Item)

Allocation A is EF1 if for all $i, j \in N$:

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}) \text{ for some } g \in A_j$$

Example

- Agent 1: $A_1 = \{a\}$, value $= 5$
- Agent 2: $A_2 = \{b, c\}$, Agent 1's value for $A_2 = (6+2)=8$
- Agent 1 envies Agent 2
- But removing item c: Agent 1's value for $\{b\} = 6$
- Now $v_1(A_1) = 5 \geq 2$ ✓ (EF1 satisfied)

Round-Robin Algorithm

Input: A fair allocation instance $I = \langle N, M, v \rangle$ with n agents and m goods.
Output: Allocation $A = (A_1, \ldots, A_n)$.
for each agent $i \in N$ **do**
    $A_i \leftarrow \varnothing$
end for
for $t = 1, \ldots, m$ **do**
    Let $i \leftarrow t \bmod n$
    Let $g^* \in \arg\max_{g \in M} v_i(g)$
    $A_i \leftarrow A_i \cup \{g^*\}$
    $M \leftarrow M \setminus \{g^*\}$
end for

return A

Property & Correctness

- Agents take turns in a cyclic order: $1, 2, \ldots, n, 1, 2, \ldots$
- Each agent picks their most valuable remaining good
- Produces EF1 (Envy-Free up to One Good) allocation
- Runs in polynomial time: $O(mn)$

Envy-Cycle Algorithm

Input: A fair allocation instance $I = \langle N, M, v \rangle$ with n agents and m goods.
Output: Allocation $A = (A_1, \ldots, A_n)$.
for each agent $i \in N$ **do**
    $A_i \leftarrow \varnothing$
end for
for $\ell = 1, \ldots, m$ **do**
    while there does not exist an unenvied agent **do**
        Find an envy-cycle $C = (i_1, \ldots, i_d)$ and resolve the cycle as follows:

$$A_i^C = \begin{cases} A_{i_{j+1}} & 1 \leq j \leq d-1 \\ A_{i_1} & j = d \end{cases}$$

        $A_i \leftarrow A_i^C$ for all $i \in C$
    end while
    Let i be an unenvied agent
    Let $g^* \in \arg\max_{g \in M} v_i(g)$
    $A_i \leftarrow A_i \cup \{g^*\}$
    $M \leftarrow M \setminus \{g^*\}$
end for

Correctness

The algorithm maintains EF1 throughout and produces EF1 allocation

Pareto Optimality

Definition (Pareto Optimal (PO))

An allocation A is Pareto optimal if there is no other allocation A' such that:

- $v_i(A'_i) \geq v_i(A_i)$ for all agents $i \in N$
- $v_j(A'_j) > v_j(A_j)$ for at least one agent $j \in N$

In other words we will say that such as allocation is not pareto dominated by any other allocation

Example

Three agents A_1, A_2, A_3 and three allocations A, B, C:

Agent	A	B	C
A_1	30	40	35
A_2	20	10	20
A_3	15	10	20

- C does not dominate B and A does not dominate B
  Therefore, B is PO
- A does not dominate C and B does not dominate C
  Therefore, C is PO

Maximum Nash Welfare (MNW)

Definition (Maximum Nash Welfare)

MNW allocation maximizes the product of utilities:

$$\text{maximize} \prod_{i \in N} v_i(A_i)$$

Equivalently: maximize $\sum_{i \in N} \log(v_i(A_i))$

Example

Two agents, two items with values:

Agent	Item 1	Item 2
1	4	1
2	1	4

- Allocation 1: $A_1 = \{1\}, A_2 = \{2\}$
  Nash welfare $= 4 \times 4 = 16$
- Allocation 2: $A_1 = \{1, 2\}, A_2 = \{\}$
  Nash welfare $= 5 \times 0 = 0$

MNW chooses Allocation 1

Properties of MNW

Property

Every MNW allocation is both EF1 and Pareto Optimal (PO)

Intuition.

- **EF1**: Maximizing product balances utilities - prevents severe envy
- **PO**: If a Pareto improvement existed, product could be increased

$\square$

MNW vs Other Solutions

Solution Concept	Fairness	Efficiency
Round-Robin	EF1	Not always PO
Envy-Cycle	EF1	Not always PO
MNW	EF1	PO
EFX	Stronger than EF1	Often PO

Key Insight

MNW provides the unique combination of EF1 + PO.

EFX Definition

Definition (Envy-Free up to Any Item)

Allocation A is EFX if for all $i, j \in N$:

$$v_i(A_i) \geq v_i(A_j \setminus \{g\}) \text{ for all } g \in A_j$$

Example

- Stronger than EF1: Must hold for **any** item removal
- Agent 1: $A_1 = \{a\}$, value $= 6$
- Agent 2: $A_2 = \{b, c\}$, Agent 1's value for $A_2 = (4+2)=6$
- Agent 1 envies Agent 2
- For EFX: Must satisfy:
  - $v_1(A_1) \geq v_1(A_2 \setminus \{b\})$
  - $v_1(A_1) \geq v_1(A_2 \setminus \{c\})$

Computation of EFX Allocation for Special Cases

Known Results

- **2 agents:** Always exists, polynomial-time
- **3 agents:** Existence proven (complex algorithm)

Special Valuation Classes

- Identical valuations
- Ordered instances
- Bi-valuations

Identical Valuation

Definition (Identical Valuation)

All agents have same valuation: $v_1 = v_2 = ... = v_n$

Leximin Allocation

1st maximize the minimum values among all bundle, then maximize the 2nd minimum value .

Leximin++ with Tie-breaking

Leximin++ with Tie-breaking by comparing the bundle size of agents in the same utility order.

Theorem

Leximin++ must produce an EFX allocation for identical valuations

Ordered Valuations

Definition (Ordered Instance)

Agents agree on item ordering: if $v_i(g) > v_i(g')$ then $v_j(g) > v_j(g')$ for all i, j

Theorem

The envy-cycle elimination algorithm computes an EFX allocation for every ordered instance

Why it works

- Common ordering simplifies envy relationships
- Stronger invariants maintained during algorithm
- Guarantees EFX instead of just EF1

EFX for Two & Three Agents

Two Agents

- EFX Always exists
- Polynomial-time
- Cut-and-choose method

Three Agents

- Existence proven
- Pseudo-polynomial time

Example

Two agents:

1. Agent 1 divides items into two bundles
2. Agent 2 chooses preferred bundle
3. Agent 1 gets remaining bundle
4. Result: EFX guaranteed

Bi-valuation

Definition (Bi-valuation)

Each item has at most two possible values:

$$v_{ij} \in \{a, b\} \text{ for fixed } a < b$$

Property

- EFX always exist.
- MNW allocation are always EFX in Bi-valuation.
- PO allocation are always EFX in Bi-valuation.

Example

Items are either "liked" or "neutral":

- Liked items: value $= 2$
- Neutral items: value $= 1$
- Not completely binary, but restricted

Future work

Research Focus

The project aims to explore the existence of **EFX (Envy-Free up to any eXcepted good)** allocation algorithms for indivisible goods under restricted graph structures such as:

- Trees
- Bipartite graphs
- Other simple graph classes

Graph-Restricted Envy

Envy can occur **only between agents who share an edge** in the graph.

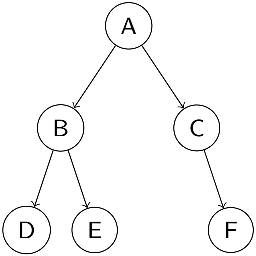

Figure: Tree Graph Example

Future work

Expected Contributions

- Determine whether **EFX allocations always exist** for these restricted graph classes.
- Design or identify **efficient algorithms** for computing EFX allocations on trees and bipartite graphs.
- Provide theoretical insights for fair division under graph-constrained settings.

Overall Expected Result

A framework and potential algorithms for computing **graph-restricted EFX allocations** for indivisible goods.

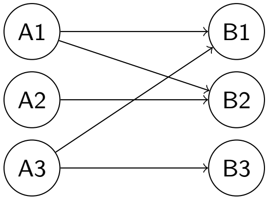

Figure: Bipartite Graph Example

Thank You!

Questions?